

Kafka Overview

1. Kafka Broker

A Kafka Broker is the Kafka server. It stores topic data, receives data from producers, and sends data to consumers. A broker typically runs on port 9092. Starting the broker:

```
kafka-server-start.bat server.properties
```

2. Kafka Producer

A Producer sends (produces) messages to Kafka topics. Example:

```
kafka-console-producer.bat --bootstrap-server localhost:9092 --topic mytopic
```

Everything typed becomes a message sent to the topic.

3. Kafka Consumer

A Consumer reads (consumes) messages from Kafka topics. Example:

```
kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic mytopic --from-beginning
```

This reads all messages from the beginning of the topic.

4. ZooKeeper

In older Kafka versions (before KRaft mode), ZooKeeper is required for coordination. It manages broker metadata, topic configurations, partition leadership, and controller election.

ZooKeeper must be started before Kafka:

```
zookeeper-server-start.bat zookeeper.properties
```

5. Kafka Without ZooKeeper (KRaft Mode)

Newer Kafka versions (3.x and above) use KRaft mode, which removes the need for ZooKeeper. Kafka handles metadata internally. Only the Kafka server needs to be started after formatting the storage directory.

Summary:

- Broker: Kafka server storing and serving messages
- Producer: Sends messages to Kafka

- Consumer: Reads messages from Kafka
- ZooKeeper: Required only in older Kafka versions for coordination
- KRaft Mode: Newer Kafka runs without ZooKeeper